TRAUMA · NEET-SS / FET / INI-SS

Distal radius and carpus

1. Distal radius fractures

Normal radiographic parameters

Parameter	Normal	Acceptable after reduction (active adult)
Radial height (length)	About 11-12 mm	Shortening **under 3-5 mm**
Radial inclination	About 22-23 degrees	**Over about 15 degrees**
Volar (palmar) tilt	About 11 degrees (range 0-20)	**Dorsal tilt under 10 degrees** (neutral or better)
Ulnar variance	Neutral (about 0 ±2 mm)	**Under +2-3 mm** ulnar positive
Articular step-off / gap	0	**Under 2 mm**

- **Teardrop angle** (lateral) about 70 degrees; **AP distance** of the lunate facet.
- **Instability predictors (Lafontaine): dorsal angulation over 20 degrees, dorsal comminution, intra-articular involvement, associated ulnar fracture, age over 60**. Three or more = likely loss of reduction in a cast. Also initial **shortening over 5 mm**.

Eponyms and classifications

- **Colles':** extra-articular, **dorsal angulation and displacement**, radial shortening ("dinner-fork"), fall on outstretched hand.
- **Smith's: volar** angulation ("reverse Colles"; garden-spade deformity); **unstable**: needs **volar buttress plate**.

- **Barton's: intra-articular rim fracture with carpal subluxation** (volar Barton commoner): **buttress plate**.
- **Chauffeur's: radial styloid fracture** (associated with scapholunate injury/perilunate).
- **Die-punch:** depressed **lunate facet** fracture.
- **Frykman** (I-VIII: radiocarpal/DRUJ involvement, ulnar styloid), **Fernandez** (mechanism-based I bending, II shearing, III compression, IV avulsion, V combined), **AO/OTA 23 A/B/C**, **Melone** (four-part intra-articular).
- **Volar lunate facet ("critical corner") fragment:** small and distal; if not captured, **volar carpal subluxation** follows; needs plate placement distal to the watershed line or a fragment-specific plate/hook.

Management

- **Undisplaced or acceptable alignment:** cast or splint 4-6 weeks; check radiographs weekly for 2-3 weeks.
- **Displaced: closed reduction** (haematoma block, Bier block) and cast; **re-displacement** is common in unstable patterns.
- **Operative indications (young/active):** failed closed reduction, **unstable patterns**, articular step-off over 2 mm, **volar shearing (Smith's, Barton's)**, open fractures, associated carpal injuries, bilateral.
- **Implants: volar locking plate** (most common; **flexor carpi radialis (modified Henry) approach**; plate **proximal to the watershed line** to avoid **FPL rupture**), K-wires (DRAFFT trial: no difference between K-wires and volar plating at 12 months in dorsally displaced fractures), dorsal plates (dorsal comminution, dorsal shear; extensor irritation), fragment-specific fixation, bridging plates (dorsal spanning) and external fixation (comminuted, polytrauma).
- **Elderly (over 65):** non-operative treatment gives **similar functional outcomes** despite worse radiographs (WRIST trial and others: volar

plating vs casting in older adults gave small early differences; **AAOS** guidance: weak evidence for surgery over 65). Volar plate for active older patients with unacceptable displacement.

Complications

- **Median nerve dysfunction / acute carpal tunnel syndrome** (progressive symptoms: **urgent carpal tunnel release** with fixation).
- **EPL rupture** after **undisplaced** fractures (attrition at **Lister's tubercle**, ischaemia; about 3-8 weeks later): **EIP to EPL transfer**. After volar plating, **long dorsal screws** can rupture EPL/EDC (use shorter screws, skyline/dorsal tangential view).
- **FPL rupture** with volar plates placed **distal to the watershed line** (plate prominence): remove plate after union.
- **Malunion** (dorsal tilt, shortening: **corrective osteotomy with bone graft**; ulnar positive variance: **ulnar shortening osteotomy**).
- **Complex regional pain syndrome (CRPS): vitamin C 500 mg daily for 50 days** may reduce incidence (AAOS moderate evidence, though conflicting trials). Budapest criteria.
- **Stiffness**, DRUJ instability/arthritis, **ulnar impaction**, post-traumatic arthritis (articular step-off over 2 mm), median/ulnar nerve injury, tendon irritation.
- **Ulnar styloid fractures: base fractures** may be associated with DRUJ instability; fixation rarely needed (test DRUJ after radius fixation).

2. Scaphoid fractures

Anatomy

- About 80% covered with articular cartilage. **Retrograde blood supply** from the **dorsal carpal branch of the radial artery** entering at the **dorsal ridge (waist)**, supplying about 70-80% (including the **proximal pole**); the **superficial palmar branch** supplies the distal

tubercle. Hence **proximal pole fractures** have the highest **AVN and non-union** rates.

Clinical and imaging

- Fall on the outstretched hand; **anatomical snuffbox tenderness**, **scaphoid tubercle tenderness**, **pain on axial thumb compression** (combination increases sensitivity).
- **Scaphoid series** (PA in ulnar deviation, lateral, obliques). Initial radiographs miss about 20%.
- **Suspected fracture with normal radiographs: MRI** (best; sensitivity near 100%), or CT; otherwise immobilise and re-image at 10-14 days.
- **Displacement: over 1 mm**, **intrascaphoid angle over 35 degrees** (lateral, **humpback deformity**), **height-to-length ratio over 0.65**, **DISI** (radiolunate angle over 15 degrees).

Classification and treatment

Location	Frequency	Treatment
Distal pole / tubercle	About 10-20%	Cast 4-6 weeks; heals well
Waist	**About 65-80% (commonest)**	**Undisplaced: cast** (below-elbow cast without the thumb is as effective as including it) 6-10 weeks, **or percutaneous screw** (faster return to work/sport; **SWIFFT trial** 2020: **no long-term benefit of early fixation** for undisplaced/minimally displaced (under 2 mm) waist fractures; treat with cast and fix if non-union). **Displaced (over 1-2 mm): fix (headless compression screw)**
Proximal pole	About 10-20%	**Fix** (high non-union and AVN risk); **dorsal approach**

- Classifications: **Herbert** (A stable acute: A1 tubercle, A2 incomplete waist; B unstable acute: B1 distal oblique, B2 complete waist, B3 proximal pole, B4 transscaphoid perilunate; C delayed union; D non-union), **Russe** (fracture line: horizontal oblique, transverse, vertical oblique), Mayo.
- **Screw position:** central in the scaphoid (longest screw in the **central axis** gives most stability); **volar approach** for waist/distal fractures (percutaneous through the trapezium edge), **dorsal approach** for proximal pole.

Scaphoid non-union

- **Risk factors: proximal pole**, **displacement over 1 mm**, delayed diagnosis/treatment, smoking, instability.
- **Natural history: SNAC wrist (scaphoid non-union advanced collapse)**: radial styloid-scaphoid arthritis first, then capitate-scaphoid, then lunocapitate (**radiolunate joint spared**).
- **Treatment: non-union without arthritis**: ORIF with **bone graft**: **volar wedge graft (Fisk-Fernandez)** for humpback deformity; **vascularised bone graft (1,2-ICSRA, Zaidemberg; medial femoral condyle free flap)** for **AVN of the proximal pole** (absent punctate bleeding). **SNAC:** radial styloidectomy (stage I), **proximal row carpectomy** (needs a healthy capitate head and lunate fossa) or **scaphoid excision + four-corner fusion** (stage II-III), total wrist fusion.
- **Preiser disease:** idiopathic AVN of the scaphoid.

3. Other carpal fractures

- **Triquetrum** (second commonest carpal fracture): **dorsal chip** (avulsion/impaction): symptomatic treatment.
- **Hook of hamate:** golfers, baseball, racquet sports; **carpal tunnel view** or **CT**; **ulnar nerve (Guyon) symptoms**, **FDP tendon ruptures (ring/little)**; **excision** of the hook for non-union/symptoms.

- **Pisiform**, **trapezium** (ridge fractures; FCR), **capitate** (scaphocapitate syndrome: head rotated 180 degrees).
- **Lunate: Kienböck disease** (AVN of the lunate): **ulnar negative variance** association; **Lichtman staging** (I normal radiograph/MRI changes, II sclerosis, **IIIA collapse with normal carpal alignment**, **IIIB fixed scaphoid rotation (ring sign, radioscaphoid angle over 60 degrees)**, IV pancarpal arthritis). **Joint levelling (radial shortening osteotomy)** for ulnar-negative stage I-IIIA, revascularisation (4,5-ECA graft), **proximal row carpectomy or STT/SC fusion for IIIB**, wrist fusion/arthroplasty for IV.

4. Carpal instability and perilunate injuries (trauma aspect)

- **Mayfield progression (perilunate instability)** with wrist extension, ulnar deviation and intercarpal supination: **I scapholunate disruption, II lunocapitate, III lunotriquetral, IV lunate dislocation** (volar, "**spilled teacup**" on lateral).
- **Greater arc injuries** (through bones: **transscaphoid perilunate fracture-dislocation**, commonest) vs **lesser arc** (purely ligamentous).
- **Signs:** broken **Gilula's arcs** (three smooth arcs on PA), **triangular lunate** on PA, **spilled teacup** on lateral, capitate dorsal to the lunate.
- **Treatment: urgent closed reduction** (Tavernier manoeuvre), then **open reduction, ligament repair and internal fixation** (dorsal ± volar approach); acute **median nerve** symptoms need urgent decompression. Up to 25% are **missed** initially.
- **Scapholunate dissociation: Terry Thomas sign** (SL gap **over 3 mm**), **scaphoid ring (cortical ring) sign**, **scapholunate angle over 60-70 degrees** (normal 30-60), **DISI** (lunate extended). **Watson's scaphoid shift test.** Acute: repair; chronic reducible: reconstruction (e.g. three-ligament tenodesis); **SLAC wrist** progression (radial styloid, then radioscaphoid, then capitolunate; radiolunate spared).

- **Lunotriquetral injury: VISI** (lunate flexed), ballottement (Reagan) test, shuck test.

5. Distal radioulnar joint and TFCC

- **TFCC** components: articular disc, dorsal and volar radioulnar ligaments (primary DRUJ stabilisers), meniscal homologue, ECU subsheath, ulnocarpal ligaments (ulnolunate, ulnotriquetral).
- **Palmer classification: Class 1 traumatic** (1A central perforation: commonest, debride; **1B peripheral ulnar detachment** (with or without ulnar styloid fracture): repair; 1C distal ulnocarpal ligaments; 1D radial detachment), **Class 2 degenerative** (ulnar impaction, stages A-E).
- **Ulnar impaction syndrome:** ulnar positive variance, lunate/ulnar head chondromalacia; **ulnar shortening osteotomy** or wafer procedure.
- **DRUJ instability tests:** piano-key sign, ballottement in neutral, pronation and supination.

> Exam pearl: **Acceptable distal radius: radial shortening under 3-5 mm, dorsal tilt under 10 degrees, inclination over 15 degrees, step-off under 2 mm.** Elderly over 65: non-operative results are comparable. **EPL rupture after undisplaced fractures; FPL rupture after plates beyond the watershed line.**

> Exam pearl: **Scaphoid: retrograde blood supply, proximal pole AVN; MRI for occult fracture; SWIFFT: cast undisplaced waist fractures; fix displaced and proximal pole fractures.** SNAC/SLAC both **spare the radiolunate joint**.

Exam pearl: **Perilunate: Mayfield I-IV (SL to lunate dislocation); transscaphoid perilunate is the commonest greater-arc injury; spilled teacup = lunate dislocation.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.